
The Transcriptional Shockwave Carries the Partner: A Virtual-Cell Foundation Model for Synthetic-Lethal Co-Dependency Prediction Beyond Screened Genes

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Abstract

1 Synthetic lethality (SL) is a powerful route to selective cancer therapy, but the space
2 of candidate gene pairs is far too large to screen exhaustively, so computational
3 prioritization is essential. Existing predictors largely perform link prediction over a
4 curated SL interaction graph; their signal lives in graph topology, so a degree-only
5 probe can win the easiest evaluation split while accuracy collapses for genes absent
6 from the training graph — exactly the cold-start regime where new biology lies. Our
7 insight is that a gene’s perturbation transcriptome, the genome-wide “shockwave”
8 that already predicts its own dependency, should also encode its synthetic-lethal
9 partners. We first confirm this on a SynLethDB-derived K562 benchmark: with the
10 evaluation harness held fixed, adding the observed perturbation transcriptome more
11 than doubles partner-ranking quality ($\text{NDCG@10 } 0.042 \rightarrow 0.094$) when one partner
12 is anchored by a real Perturb-seq profile, though the lift vanishes when both genes
13 are unseen. Because an observed response exists only for screened genes, we then
14 propose generating it: an end-to-end framework that drives a frozen virtual-cell
15 foundation model (STATE) with a protein-language-model gene identity (ESM2)
16 through a trainable adapter, so a perturbation response — and an SL prediction —
17 can be produced for genes outside the backbone’s fixed vocabulary (83.7% of our
18 universe). We report this framework as a preliminary method: it runs end to end
19 and scores out-of-vocabulary genes, but does not yet clear a strong dependency-
20 only floor. Finally, a decomposition shows that most genome-wide co-dependency
21 signal is pan-essentiality, isolating pair-specific SL as the hard residual that a
22 generative virtual-cell model must learn. All results are scoped to K562 and to a
23 benchmark-adapter label, not a validated SL assay.

24 1 Introduction

25 Synthetic lethality (SL) — where disrupting either of two genes is tolerated but disrupting both kills
26 the cell — is one of the few mechanisms that turns a tumor’s own genetics into a selective drug
27 target, as the clinical success of PARP inhibitors in BRCA-mutant cancers demonstrates [O’Neil
28 et al., 2017, Huang et al., 2019]. The bottleneck is no longer whether SL can be exploited but which
29 pairs to pursue: the human genome admits on the order of 10^8 candidate gene pairs, far beyond what
30 combinatorial CRISPR screens can cover, and SL is context-specific, so a pair that is lethal in one
31 cellular background need not be in another. Computational prioritization is therefore a prerequisite for
32 experimental follow-up, and this paper focuses on the specific setting of ranking a gene’s candidate
33 SL partners within a single, well-characterized cellular context (K562).

34 Most existing predictors meet this need by link prediction over a curated SL interaction graph, and
35 they are strong when a test pair shares genes with the training graph [Feng et al., 2024, Wang et al.,

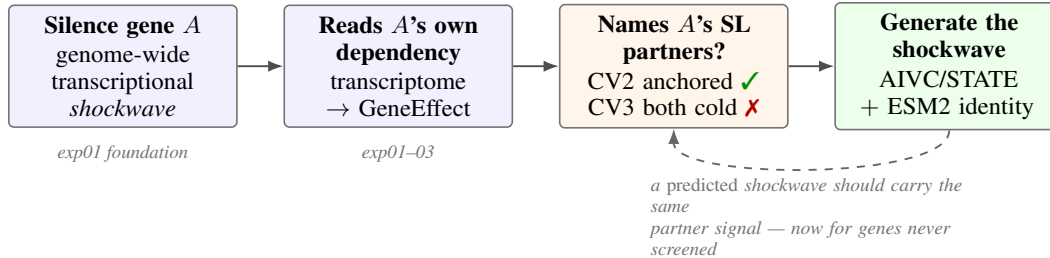


Figure 1: The transcriptional shockwave carries the partner. Perturbing a gene produces a genome-wide response that predicts the gene’s own dependency (exp01). The same observed response also ranks a gene’s synthetic-lethal partners when one partner is anchored by a real profile (CV2), but not when both genes are unscreened (CV3). Our contribution is a virtual-cell foundation model that *generates* this response from a protein-language-model gene identity, so the partner signal can in principle reach pairs neither gene was screened for (dashed loop; results preliminary).

2022b]. Their signal, however, lives in graph topology: because positive pairs are enriched among high-degree hub genes, a probe using gene degree alone can win the easiest evaluation split, while accuracy collapses once one or both genes are absent from the training graph. This exposes the real challenge — true cold-start prediction for genes with no curated interactions and, in the hardest regime, no prior measurement at all. It is precisely the regime where undiscovered biology lives, and the regime where graph-topology methods have no signal to stand on.

Our starting observation is that a different signal source sidesteps the graph entirely. When a gene is silenced, the cell’s genome-wide transcriptional response — a perturbation “shockwave” — reflects that gene’s functional wiring; the same response that predicts a gene’s own dependency [Chiu et al., 2021, Haider et al., 2025] should also carry information about the partners whose loss it cannot tolerate. We confirm this directly: with the evaluation harness held fixed, adding the observed perturbation transcriptome more than doubles partner-ranking quality in the regime where one partner is anchored by a real Perturb-seq profile (Figure 1). This is the empirical license for our approach, but it also reveals its limit — an observed response exists only for genes that were actually screened, so it cannot reach the cold-start frontier.

We therefore propose generating the response instead of measuring it. Our second contribution is an end-to-end framework that drives a frozen virtual-cell foundation model (STATE; Adduri et al., 2025) with a protein-language-model gene identity (ESM2; Lin et al., 2023) through a trainable adapter, so that a perturbation response — and hence an SL prediction — can be produced for any gene, including the 83.7% of our benchmark universe that lies outside the backbone’s fixed vocabulary. The insight is that if the *observed* shockwave encodes a gene’s partners, a faithfully *generated* shockwave should carry the same signal while extending it past the set of screened genes. We report this framework as a preliminary method: it runs end to end and produces predictions for out-of-vocabulary genes, but its current results do not yet clear a strong dependency-only baseline, and we are explicit about this throughout.

Our contributions are: (i) a leakage-controlled evaluation on a SynLethDB-derived K562 benchmark that separates an honest cold-start surface from a degree-gameable diagnostic, together with a dependency-only floor on it; (ii) the proof-of-concept that the observed perturbation transcriptome encodes a gene’s SL partners when one partner is anchored; (iii) a virtual-cell framework that generates responses for unscreened genes via an ESM2-conditioned adapter to a frozen backbone, reported with preliminary results; and (iv) a decomposition showing that most genome-wide co-dependency signal is pan-essentiality, isolating pair-specific SL as the hard residual the generative approach must learn. All claims are scoped to K562 and to a benchmark-adapter label, not a validated SL assay.

2 Related Work

Computational synthetic-lethality prediction. Most SL predictors treat the problem as link prediction over a known SL interaction graph curated in databases such as SynLethDB [Wang et al.,

2022a] and SLKB [Gokbag et al., 2023]. Matrix-factorization and graph-neural-network methods (for example the GRSMF, SL2MF, and DDGCN families surveyed by Wang et al. [2022b]) propagate observed SL labels through gene–gene similarity or interaction topology, and the unified benchmark of Feng et al. [2024] shows they perform strongly when test pairs share genes with the training graph. Their common weakness is the same as their strength: because the signal lives in graph topology and node degree, accuracy degrades sharply when one or both genes are unseen, and a degree-only probe can top the easiest split (our Table 2). We use that benchmark and its difficulty splits as our evaluation harness, but our features come from a gene’s *perturbation response* rather than its position in the SL graph, which is what gives the approach a path to genes with no curated interactions.

Virtual-cell and perturbation-response models. A second line of work models the cell’s response to perturbation directly. The virtual-cell program [Bunne et al., 2024] and the associated challenge [Roohani et al., 2025] frame this as learning to predict how a cell-state distribution shifts under genetic perturbation; STATE [Adduri et al., 2025] is a transformer that predicts such set-level responses, and single-cell foundation models such as scGPT [Cui et al., 2024] and Tahoe-x1 [Gandhi et al., 2025] pretrain transferable embeddings at scale. Two limitations matter for our setting. First, these models identify a perturbation through a fixed gene vocabulary, so out-of-vocabulary genes cannot be queried without an adapter — the bottleneck our ESM2-conditioned adapter [Lin et al., 2023] targets. Second, careful benchmarking finds that deep perturbation predictors do not yet consistently beat simple linear baselines [Ahlmann-Eltze et al., 2025, Vinas Torne et al., 2025], which is why we frame our generative framework as a preliminary method and compare it candidly against a strong dependency-only floor rather than claiming superiority.

Cancer dependency maps and the essentiality–SL distinction. A third line characterizes single-gene dependencies from genome-scale CRISPR screens. The DepMap effort, using CERES/Chronos correction [Meyers et al., 2017], quantifies how much each gene’s loss impairs fitness, and downstream work builds clinically informed and translational dependency maps and target-prioritization frameworks [Pacini et al., 2024, Shi et al., 2024], including deep-learning prediction of dependency from tumor omics [Chiu et al., 2021]. We rely on these dependency scores as features and as a prediction target for a gene’s own essentiality, but we are careful not to conflate essentiality with synthetic lethality: a broadly essential gene is not an SL partner, and our decomposition shows that genome-wide co-dependency signal is dominated by pan-essentiality rather than pair-specific interaction. The observation closest to our hypothesis is that tumor transcriptomic architecture reflects SL interactions [Haider et al., 2025]; we make the predictive direction explicit and, crucially, generate the transcriptome rather than requiring it to have been measured.

3 Method

We cast synthetic-lethal (SL) candidate prioritization as ranking gene pairs by a predicted SL probability, where the features of each gene are derived from its *generated* perturbation response rather than from a pre-existing screen. Our central design choice is to obtain that response from a frozen virtual-cell foundation model driven by a protein-language-model gene identity, so that the pipeline can in principle score a gene that was never perturbed in the laboratory. Figure 2 shows the forward path. Section 3.1 fixes the prediction target and ranking protocol; Section 3.2 introduces the perturbation adapter that breaks the foundation model’s closed vocabulary; Section 3.3 describes how a frozen backbone turns an adapted identity into a predicted response bag; Section 3.4 defines the swap-invariant pair head; and Section 3.5 gives the three-part training objective and the evaluation protocol, including an honest statement of how model selection interacts with the benchmark.

3.1 Problem setup and prediction target

We work on the K562-filtered, SynLethDB-derived benchmark of Feng et al. [2024], whose gene universe contains 9,471 genes that carry numeric K562 DepMap GeneEffect values. A training example is an unordered gene pair (a, b) with a binary benchmark label $D_{ab} \in \{0, 1\}$ from a balanced Rand 1:1 split. For each gene g we also have its DepMap dependency scalar $C_g = \text{GeneEffect}(\text{K562}, g)$, a population-level fitness readout. The model outputs a probability $P(\text{SL} \mid a, b)$, and we evaluate it with the benchmark’s per-anchor protocol: for each anchor gene we rank its candidate partners and average ranking quality (NDCG@ k , MAP@ k) across anchors, alongside

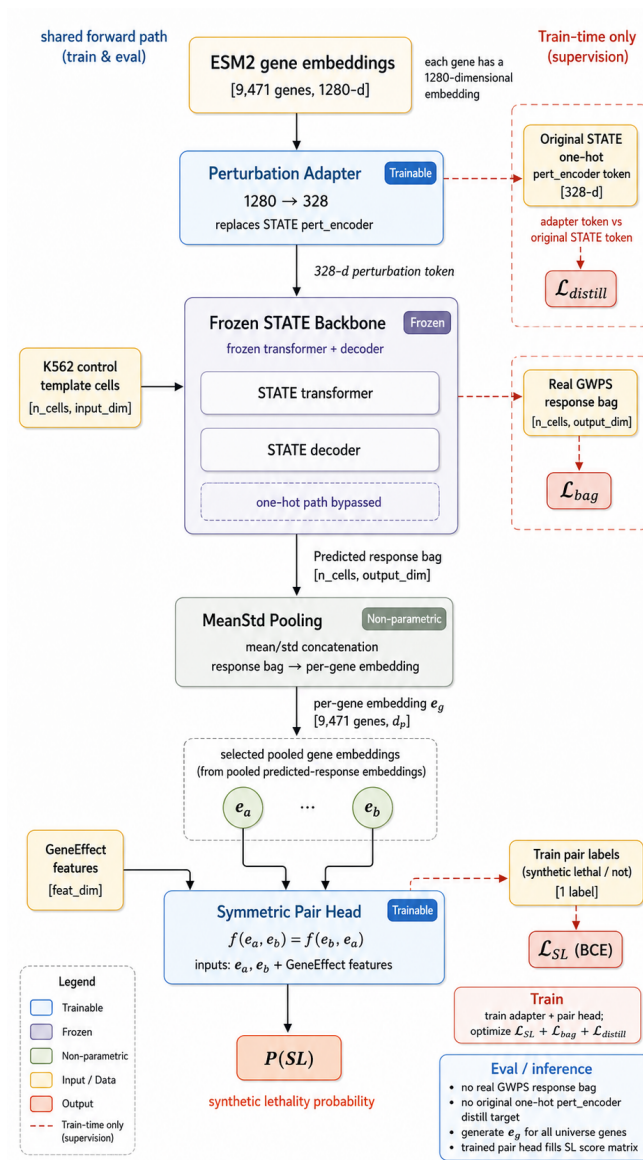


Figure 2: The end-to-end framework. A gene’s ESM2 identity embedding is mapped by a trainable perturbation adapter into the perturbation space of a frozen STATE backbone, which generates a predicted response bag on a control-cell template. Mean–std pooling yields a gene embedding; a swap-invariant pair head produces the SL probability. Only the adapter and pair head are trainable (blue); the backbone is frozen (snowflake). Training combines an SL cross-entropy loss with a token-distillation loss and real-bag supervision.

threshold-free classification (AUROC, AUPR). We report three splits of increasing cold-start severity: CV1 (random pair-level holdout), CV2 (one gene of each test pair held out of training), and CV3 (both genes held out). Because D is a benchmark-adapter label rather than a validated K562 SL assay, and its negatives are unconfirmed non-SL pairs, we treat all outputs as candidate prioritization, not target validation.

3.2 Perturbation adapter: breaking the closed vocabulary

The obstacle to scoring unscreened genes is that our virtual-cell backbone, Arc Institute’s STATE model [Adduri et al., 2025], identifies a perturbation through a one-hot lookup over the fixed set of

genes it was trained on. In our checkpoint that vocabulary covers only 1,542 of the 9,471 benchmark genes (16.3%), so a one-hot interface alone can never produce a response for the remaining 83.7% of the universe. We therefore replace the one-hot perturbation input with a learned adapter that maps a continuous, open-vocabulary gene identity into the backbone’s perturbation space.

We represent every gene by its ESM2 protein-language-model embedding [Lin et al., 2023], a 1280-dimensional vector available for any protein-coding gene regardless of whether it was ever screened. The *perturbation adapter* is a two-layer multilayer perceptron, $\text{Linear}(1280 \rightarrow 512) \rightarrow \text{GELU} \rightarrow \text{Linear}(512 \rightarrow d_p)$, whose output width d_p matches STATE’s native perturbation-token dimension. Given a gene’s ESM2 vector, the adapter emits a perturbation vector that is then consumed by STATE’s own (frozen) perturbation encoder, exactly where the one-hot token would have entered. The advantage is reach: because the adapter’s input is a smooth function of protein identity rather than a discrete index, the same trained module yields a perturbation token for genes the backbone never saw, which is the prerequisite for cold-start (CV3) scoring.

3.3 Generating the perturbation response with a frozen backbone

Given the adapted perturbation token, we generate the gene’s response on a fixed template of 256 K562 control cells: STATE maps (control cells, perturbation token) to a predicted post-perturbation expression bag of the same shape, our in-silico analogue of a Perturb-seq readout. The entire STATE backbone is frozen (all parameters held at `requires_grad = False` and kept in evaluation mode); only the adapter and the downstream pair head are trained. Gradients still flow into the adapter through the frozen perturbation encoder, so the adapter learns to drive the backbone toward useful responses without disturbing the pretrained dynamics. Freezing is a deliberate data-efficiency choice: the benchmark provides only tens of thousands of labeled pairs, far too few to fine-tune a foundation model, whereas a small adapter and head are learnable from this signal while inheriting the backbone’s pretrained perturbation biology.

3.4 Symmetric pair head

Synthetic lethality is an unordered relation, so the scorer must satisfy $f(a, b) = f(b, a)$; a head that treats (a, b) and (b, a) differently would learn spurious order artifacts. We first pool each predicted response bag into a gene embedding e_g by concatenating the per-feature mean and standard deviation across the 256 cells, which summarizes both the average shift and the dispersion of the generated response. We then combine the two gene embeddings through three element-wise swap-invariant blocks — the sum $e_a + e_b$, the absolute difference $|e_a - e_b|$, and the product $e_a \odot e_b$ — and append a small block of swap-invariant DepMap GeneEffect features (symmetric functions of C_a and C_b) together with a coverage indicator. A multilayer perceptron maps this representation to the pair logit, whose sigmoid is $P(\text{SL} \mid a, b)$. Every operation upstream of the logit is symmetric in a and b , so the head is exactly swap-invariant by construction rather than by augmentation.

3.5 Training objective and evaluation protocol

The adapter and head are trained with a three-part objective that supervises the pipeline at the levels where signal is available. The primary term is a binary cross-entropy loss on the pair logit against the benchmark label D_{ab} . A *distillation* term keeps the adapter compatible with the pretrained backbone: for the in-vocabulary genes it minimizes the squared error between the adapter-produced perturbation token and STATE’s own one-hot token, anchoring the open-vocabulary embedding to the space the backbone expects. A *bag supervision* term aligns the generated response with reality for genes that do have a real profile, minimizing the discrepancy (mean shift plus an energy distance) between the predicted and observed genome-wide Perturb-seq bags. To avoid leakage, bag supervision is applied only to training genes with a real profile; held-out genes are never shown a real bag. A short warmup emphasizes the distillation and bag terms before the SL term is activated, and the combiner drops any zero-weighted non-finite term so that an inactive loss cannot destabilize the gradient.

We deliberately state one limitation of the current evaluation. Because the benchmark protocol selects the best training epoch using the test-fold metric, our reported numbers are selection-matched to that protocol and are therefore not a strict embedding-only ablation against the dependency-only baseline. We report this openly and, in the experiments, treat the resulting numbers as preliminary single-fold status rather than as a completed cross-validated comparison.

Table 1: Foundation: a gene’s perturbation transcriptome predicts its *own* DepMap dependency, and the signal is real transcriptomic structure rather than a generic death axis. exp01 pseudobulk Δ -expression $\rightarrow$ GeneEffect; exp02 audit against the death signature; exp03 single-cell distribution embeddings (exploratory). The exp03 Adamson number came from an Adamson-guided sweep and is not a held-out generalization result.

Setting	Metric	Value
exp01 Replogle K562 (5-fold CV)	Spearman $\uparrow$	0.485
exp01 Adamson same-line transfer	Spearman $\uparrow$	0.500
exp01 Adamson same-line transfer	AUROC $\uparrow$ (GE < -1)	0.886
exp02 death signature only	Spearman $\uparrow$	0.244
exp02 best pseudobulk transcriptome	Spearman $\uparrow$	0.494
exp02 death-residualized transcriptome	Spearman $\uparrow$	0.503
exp03 scVII128 GMM-Ridge (Adamson sweep †)	Spearman $\uparrow$	0.666
exp03 scVII128 GMM-Ridge (Adamson sweep †)	AUROC $\uparrow$	0.911

† Adamson-guided sweep; exploratory, not held-out.

185 4 Experiments

186 We organize the evidence around the paper’s logic: first that a perturbation transcriptome predicts
 187 a gene’s own dependency at all (the foundation), then that the observed transcriptome encodes a
 188 gene’s SL partners when one partner is anchored (the proof-of-concept that licenses the method),
 189 then the current status of the generative framework, and finally a decomposition that explains why
 190 the cold-start residual is hard.

191 4.1 Setup

192 All SL experiments use the K562-filtered, SynLethDB-derived benchmark of Feng et al. [2024] (a
 193 9,471-gene universe, balanced Rand 1:1 pairs) under its per-anchor `cal_metrics` ranking protocol,
 194 reporting AUROC, AUPR, and NDCG@ k /MAP@ k . We use three splits of increasing cold-start
 195 severity: CV1 (random pair-level holdout), CV2 (one gene of each test pair held out), and CV3 (both
 196 genes held out). CV1 is reported only as a diagnostic: because positives and negatives differ in node
 197 degree, a probe using gene degree alone tops it (Table 2), so CV1 cannot demonstrate generalization
 198 to unseen genes. Generalization claims therefore come only from CV2 and CV3. The label D is
 199 a benchmark-adaptor target whose negatives are unconfirmed non-SL pairs; we never interpret a
 200 high-ranked pair as a validated K562 SL target.

201 4.2 Foundation: the perturbation transcriptome predicts a gene’s own dependency

202 Before asking whether a perturbation response names a gene’s partners, we establish that it predicts
 203 the gene’s *own* dependency (Table 1). Pseudobulk Δ -expression regressed onto DepMap GeneEffect
 204 reaches a 5-fold cross-validated Spearman of 0.485 on Replogle K562, and a same-cell-line transfer
 205 to the Adamson screen reaches AUROC 0.886 for strong dependencies; we treat the Adamson transfer
 206 as a same-line gate, not broad validation. An audit rules out the trivial explanation that the model
 207 only reads a generic cell-death axis: the death signature alone reaches Spearman 0.244, far below the
 208 transcriptome’s 0.494, and residualizing the death signature out of the transcriptome leaves the signal
 209 essentially intact at 0.503. Single-cell distribution embeddings carry the same signal across datasets
 210 (exp03), though we flag that the strongest Adamson number there came from an Adamson-guided
 211 sweep and is exploratory, not a held-out result. Together these establish that the perturbation response
 212 encodes real, dependency-relevant transcriptional structure.

213 4.3 A dependency-only floor, and a degree-gameable diagnostic

214 We next fix the bar that any transcriptome-aware model must clear: a dependency-only classifier built
 215 from swap-invariant DepMap GeneEffect features (Table 2). On the honest splits this floor is already
 216 strong — CV2 AUROC 0.704 — which sets a demanding reference. The table also makes the CV1

Table 2: Dependency-only floor on the K562 SL benchmark (exp06), per-anchor `cal_metrics` ranking. CV1 (pair-level) is degree-gameable: a degree-only probe (shaded) wins it, so CV1 ranking is reported as a diagnostic, not as generalization. CV2 (one gene held out) and CV3 (both genes cold) are the honest surfaces.

Model	CV1 (diagnostic)			CV2 (one held out)			CV3 (both cold)	
	AUROC↑	AUPR↑	NDCG@10↑	AUROC↑	AUPR↑	NDCG@10↑	AUROC↑	AUPR↑
Logistic reg. (Model A)	0.621	0.648	0.004	0.620	0.648	0.005	0.617	0.648
XGBoost (Model B)	0.795	0.812	0.050	0.704	0.732	0.042	0.596	0.604
Degree probe (Model C)	0.849	0.902	0.197	0.500	0.750	0.001	0.500	0.750

Table 3: Adding the observed perturbation transcriptome to the dependency-only features (exp07; identical harness, seeds, and splits, only the feature matrix changes). On CV2, where one partner is anchored by a real Perturb-seq profile, transcriptome features more than double ranking quality (NDCG@10 row, shaded). On CV3 (both genes cold, no observed profile) the lift vanishes. `full_universe` = all benchmark pairs; `covered_pairs` = pairs whose anchored gene has a real profile.

Split	Features	AUROC↑	NDCG@10↑	MAP@10↑
CV2	dependency-only	0.704	0.042	0.034
CV2	+ transcriptome	0.751	0.094	0.079
CV3	dependency-only	0.596	0.002	0.001
CV3	+ transcriptome	0.630	0.001	0.001

217 caveat concrete: a degree-only probe wins CV1 ranking (NDCG@10 0.197) yet collapses to 0.001
 218 on CV2 and CV3, confirming that CV1 rewards topology rather than biology. We therefore read CV1
 219 as a sanity diagnostic and judge every subsequent result on CV2 and CV3.

220 4.4 Proof-of-concept: the observed transcriptome encodes SL partners

221 The pivotal question is whether the observed perturbation transcriptome carries partner information
 222 beyond what dependency scores already provide. Holding the harness, seeds, and splits fixed and
 223 changing only the feature matrix, adding the observed transcriptome more than doubles CV2 ranking
 224 quality (NDCG@10 0.042 → 0.094; AUROC 0.704 → 0.751; Table 3), and the lift concentrates on
 225 the covered-pair slice where real profiles exist. This is the empirical license for the whole approach:
 226 when one partner is anchored by a real response, that response encodes its synthetic-lethal partners.
 227 The same table delivers the matching bad news. On CV3, where neither gene has an observed profile,
 228 the transcriptome lift vanishes (Figure 1 and the difficulty ladder, Figure 3). Cold-start ranking is the
 229 open frontier, and it is exactly the regime an observed-only method cannot reach, because there is no
 230 shockwave to read.

231 4.5 The generative framework: preliminary status

232 We now report the current status of the framework that targets the CV3 frontier by *generating*
 233 responses for unscreened genes (Table 4). These results are preliminary. The runs are incomplete
 234 (single selected folds, not a 5-fold mean), the best-epoch selection reads the test fold (Section 3.5),
 235 and the best folds do not yet clear the dependency-only floor (CV2 AUROC 0.667 versus the floor’s
 236 0.704). We make no claim that the framework beats any baseline; we report it as the present state of
 237 a newly proposed method whose tuning is ongoing. What the experiment does establish is that the
 238 pipeline runs end to end and produces calibrated SL probabilities for genes outside STATE’s 16.3%
 239 in-vocabulary set, which an observed-transcriptome method structurally cannot do.

240 4.6 Decomposition: most cold-start signal is pan-essentiality

241 Finally we explain why the cold-start residual is so hard (Table 5). Cross-cell-line selectivity features
 242 lift classification on both honest splits (CV2 AUROC 0.704 → 0.742, CV3 0.596 → 0.645), which

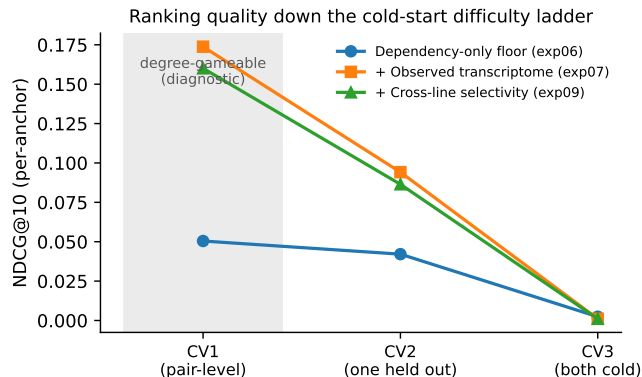


Figure 3: Ranking quality down the cold-start difficulty ladder. The observed transcriptome lifts CV2 (one gene anchored) but every signal — dependency, transcriptome, and cross-line selectivity — collapses on CV3 (both genes cold). CV1 is shaded as the degree-gameable diagnostic.

Table 4: **Preliminary** results for the virtual-cell framework (exp08, AIVC/STATE+ESM2 adapter). Values are the *best selected fold*, not a 5-fold mean, and best-epoch selection reads the test fold; full 5-fold tuning is ongoing (see Limitations). The dependency-only floor (exp06) is shown only for reference. We make no claim that the framework beats the floor; these numbers report the current status of a newly proposed method.

Split	Method	AUROC↑	NDCG@10↑
CV2	dependency-only floor (ref.)	0.704	0.042
CV2	AIVC/STATE+ESM2 (best fold, prelim.)	0.667	0.005
CV3	dependency-only floor (ref.)	0.596	0.002
CV3	AIVC/STATE+ESM2 (best fold, prelim.)	0.587	0.003

might suggest progress on co-dependency. But when the benchmark is restricted to non-pan-essential pairs on CV3, performance falls back toward chance (AUROC 0.583, AUPR 0.490). The lift was therefore carried mostly by pan-essentiality structure — genes that are broadly essential are easy to flag — while pair-specific synthetic lethality, the biologically interesting relation, remains the thin and unsolved residual. This reframes the cold-start gap as a precise target: a generative virtual-cell model must learn the pair-specific signal that genome-wide essentiality gravity hides, which is exactly what the framework in Section 3 is designed to pursue.

5 Conclusion

We studied whether a gene’s perturbation transcriptome encodes its synthetic-lethal partners, and how to extend that signal to genes that were never screened. On a SynLethDB-derived K562 benchmark with a leakage-controlled difficulty ladder, we showed that the observed perturbation transcriptome more than doubles partner-ranking quality when one partner is anchored by a real Perturb-seq profile, establishing that the same shockwave which reads a gene’s own dependency also carries information about its partners. To reach genes with no observed response, we proposed an end-to-end framework that generates the response from a frozen virtual-cell foundation model conditioned on a protein-language-model gene identity, breaking the backbone’s closed vocabulary. A decomposition then clarified why the cold-start residual is hard: most genome-wide co-dependency signal is pan-essentiality, leaving pair-specific synthetic lethality as a thin residual that a generative model must isolate.

Several limitations bound these claims. The generative framework is preliminary: its current runs are incomplete and do not yet clear the dependency-only floor, so we make no superiority claim. All results are for a single cellular context (K562) and a benchmark-adaptor label whose negatives are unconfirmed non-SL pairs, not a validated K562 SL assay, so high-ranked pairs are candidates for follow-up rather than confirmed targets. Cold-start ranking (both genes unseen) remains unsolved

Table 5: Cross-cell-line selectivity features (exp09). They lift classification on both honest splits, but when the benchmark is restricted to non-pan-essential pairs on CV3 (shaded), AUROC and AUPR collapse toward chance. Most of the genome-wide co-dependency signal is pan-essentiality structure; pair-specific synthetic lethality is the thin, hard residual.

Split	Features	AUROC↑	AUPR↑	NDCG@10↑
CV2	dependency-only	0.704	0.732	0.042
CV2	+ selectivity	0.742	0.769	0.086
CV3	dependency-only	0.596	0.609	0.002
CV3	+ selectivity	0.645	0.651	0.001
CV3	+ selectivity, non-pan-ess.	0.583	0.490	0.001

by every signal we tested. The natural next steps are to complete cross-validated tuning of the generative framework with an evaluation that does not select on the test fold, to extend beyond K562 to cross-lineage and patient-context transfer, and to supervise the model directly on the pair-specific residual that pan-essentiality otherwise hides.

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370 **A Artifact and reproducibility notes**

371 All quantitative values in the tables are extracted directly from on-disk experiment artifacts by a
 372 single tested script (`docs/report/scripts/make_tables.py`); no number is entered by hand.
 373 The benchmark splits, per-anchor ranking protocol, and feature definitions follow the experiment
 374 logs referenced in each table caption.